

AI Course

Capstone Project Action Plan

For students (instructor's review required)

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Course	AI Course
Team Name	Cogni
Team Leader/ Members	Marwan Elamami (Team Leader) / Abdulrahman Khengari, Ahmed Alaeb, Sanad Ali, Mohammed Thabet, Abdul Majid
Project Title	Bayan: An AI Reading Assistant for Arabic Readers with Dyslexia
Goal	• Help Arabic readers with dyslexia read and understand Arabic text on their own. • Simplify complex Arabic text automatically without changing its meaning. • Add full diacritics (tashkeel) so every word can be pronounced correctly. • Read the text aloud with synchronized word-by-word highlighting. • Deliver this inside the apps people already read in: selecting Arabic text anywhere on the phone shows “تبسيط” (Simplify) in the text-selection menu. • Run every model on the device, so the reader's text never leaves the phone and the tool works offline.
Abstract	Arabic creates specific barriers for readers with dyslexia: short vowels are usually omitted, prefixes and suffixes attach directly to words, and long sentences chain many clauses together. Most dyslexia tools target Latin-script languages; we found no deployed Arabic dyslexia-support software. Bayan is a three-stage on-device pipeline that simplifies complex Arabic text with a fine-tuned Arabic encoder-decoder Transformer, restores its diacritics, and reads it aloud with synchronized word highlighting. It is delivered as an Android plugin registered on the system PROCESS_TEXT intent, so “تبسيط” (Simplify) appears in the text-selection menu of every app. The reader is helped where the difficulty occurs — no copying, no pasting, no separate app to open. Every claim below is labelled measured, decided or planned. Measured to date: a copy-the-input baseline on two locked test sets, a first simplification model scored against it, a four-model diacritization benchmark, and 9.3% test-set contamination detected and removed from a published Arabic simplification corpus before any training use.
Method	1. Data construction (planned, in progress): easy BAREC sentences and religious scripture are kept unchanged. For each hard sentence an LLM generates several candidate simplifications; an LLM judge calibrated on human annotations scores meaning preservation, and a fine-tuned Arabic BERT classifier checks readability. Every LLM-generated pair passes a human-reviewed quality gate before it can be used for training, and the source text, judge score, reasoning and readability label are stored with each accepted pair as its validation evidence. 2. Simplification (planned): fine-tune Arabic encoder-decoder Transformers — AraT5v2-base-1024 (368M parameters) and AraBART (139M) — on identical data and compare them before choosing one to compress. A first AraT5v2 model is trained and scored; the AraBART run has not started, and the choice between them will be made on measured results. Training uses

level tokens so a single model offers three simplification strengths plus a “leave unchanged” mode, because knowing when not to simplify is part of the task.

3. Diacritization (decided, measured): Libtashkeel, selected from a four-model benchmark on the Tashkeela test split — DER 6.91% excluding case endings, 4.79 MB, 125 sentences/sec, MIT licence, exports to ONNX. CAMEL Tools was rejected on a harder ground than accuracy: it altered base letters in 32% of sentences. Diacritization error and base-letter change rate are re-measured on our own held-out simplified output, not only on raw corpus text.
4. Read-aloud (decided): a pretrained Arabic text-to-speech model with word timestamps for synchronized highlighting.
5. Evaluation (measured and planned): every score is reported beside a copy-the-input baseline on the same split, because on a corpus containing many already-simple sentences a model that does nothing can score well. Test sets are locked and recorded by SHA-256 in a committed manifest, and a leakage check runs over every training file before it is used. Automatic metrics: SARI, BLEU, readability reduction measured by an independent BAREC model, BERTScore, and DER/WER. Human evaluation with Arabic readers who have dyslexia covers comprehension, reading effort and acceptability (see Expected Outcome).
6. Application (decided): one Android plugin on the PROCESS_TEXT intent with a Material 3 bottom sheet, ONNX Runtime Mobile for every model, greedy decoding, and the model downloaded on first run to keep the install small. On-device latency, model size and quality are measured on a real phone and reported together as a trade-off, not separately.

Data	Sources and size • BAREC (CAMEL Lab, CC BY-SA 4.0): 69,441 sentences with readability labels — source sentences to simplify, and the labels for the readability classifier. • SAMER Simplification Corpus v1 (CAMEL Lab): 14,343 complex-to-simple training pairs at three levels. CAMEL Lab approved fine-tuning and publishing the resulting weights non-commercially; the corpus itself is never redistributed. • Baseet (University of Jeddah): 34,294 sentences with three LLM-generated levels. 3,190 rows (9.3%) overlapped our locked test sets and were removed before any training use; 31,104 remain. • DAASI (CCO): 1,509 human-written pairs of government and insurance text, plus a 350-pair held-out test set — our only out-of-domain test set with human-written references. • Bayan corpus (built by the team): LLM-generated pairs behind a human-reviewed quality gate; 5,585 generated so far, target at least 5,000 accepted. • Tashkeela: fully diacritized Arabic text for validating the diacritization stage. • Training mix: approximately 107,000 pairs after filtering, across four corpora and three domains (literary, general, government forms). Pair-selection criteria A generated pair is accepted only if the judge scores it meaning-preserving, the readability classifier confirms the simplified side is easier, the base letters are unchanged, and a human reviewer approves it. Sentences already easy, and religious scripture and poetry, are kept unchanged and used as “leave unchanged” training examples. Annotator agreement Two annotators independently label an overlapping sample of 200 pairs; agreement is reported as Cohen's kappa, with disagreements adjudicated by the Data Lead and the guideline revised where they cluster. Owner: Abdulrahman Khengari, Week 3. Leakage-safe split Two splits are locked before any training and never trained on: BAREC test (7,286 rows) and SAMER test (3,277 rows), each recorded by row count and SHA-256 in a committed manifest so anyone can verify they hold the same data. Every training file is checked against both for exact and near-duplicate matches before use; this is what found the 9.3% overlap in Baseet.
Expected Outcome	Deliverables • A working Android plugin: the reader selects Arabic text in any app, taps “تبسيط” (Simplify), and receives a simplified, fully diacritized version, read aloud with synchronized word highlighting. • Fine-tuned simplification model(s) quantized for on-device use, benchmarked against a copy-the-input baseline, a published system and large language models. • A new Arabic simplification corpus, an evaluation benchmark, and a reusable test-set leakage check. Results to date (measured) • Copy-the-input baseline on both locked test sets: SARI 77.5 overall on SAMER test, but 56.1 on the rows a human actually simplified — the bar any model must clear. • First model fine-tuned on SAMER and evaluated. It does not yet clear that baseline under the greedy decoding we ship on-device; the measured gap set the recipe for the next runs. • Diacritization benchmark across four models: DER, WER, base-letter change rate, size, speed

and licence.

- 9.3% test-set contamination found and removed from a published Arabic corpus before use.

Human evaluation

Twelve to fifteen Arabic readers with dyslexia, recruited through schools and local associations, each read matched original and simplified passages. We measure comprehension (questions answered correctly), reading effort (self-reported, five-point scale) and acceptability. Participation is under written informed consent, responses are anonymised at collection, and no personal data is stored with them. Owners: Abdul Majid and Marwan Elamami, Week 3.

On-device trade-offs

Latency per sentence, model size and quality are measured together on a real mid-range phone and reported as one trade-off. Size budget for the simplifier is 250 MB: int8 after vocabulary pruning is about 223 MB, with int4 (about 112 MB) as the fallback if load time or memory is the constraint. Latency is left blank until it is measured, rather than estimated.

Benefits

- Greater reading independence and comprehension for Arabic readers with dyslexia.
- Support for students, teachers and schools, and for readers with low literacy and learners of Arabic.
- Everything runs on the device, so it works offline and the reader's text stays private.

Role by Member

Marwan Elamami | [Team Leader & On-Device Model Lead](#)

Project management; system architecture; model compression and quantization for on-device use; integration; human evaluation with dyslexic readers.

Abdulrahman Khengari | [Data Lead](#)

Building and validating the simplification corpus; readability classifier; annotation guidelines and annotator agreement.

Ahmed Alaeb | [Model Training Lead](#)

Training and evaluating the simplification models; corpus conversion; training infrastructure.

Sanad Ali | [Machine Learning Engineer](#)

Diacritization model selection and benchmarking; co-training the simplification models.

Mohammed Thabet | [Application Engineer](#)

Android plugin development; on-device inference integration; read-aloud with word highlighting.

Abdul Majid | [Evaluation & Benchmarking Lead](#)

Test-set design and locking; leakage detection; the scoring pipeline; automatic and human evaluation.

Schedule Summary

- Week 1 (Wed 16 – Sun 20 Sep) — complete: data pipeline and validators; readability classifier; test-set design; product-form decision; application skeleton.
- Week 2 (Mon 21 – Sun 27 Sep) — in progress: corpus generation and review; first and second simplification training runs; diacritization benchmark; locked test sets and scoring pipeline; Android plugin with on-device inference.
- Week 3 (Mon 28 Sep – Sun 4 Oct): architecture frozen; int8 quantization with size and latency measured on a real device; annotator agreement reported; full benchmark including human error annotation; human evaluation with dyslexic readers; end-to-end integration.
- Week 4 (Mon 5 – Tue 6 Oct): final report and demo video for submission on Mon 5 Oct; final pitch on Tue 6 Oct.

Detailed tasks, owners and daily status are tracked in the Work Breakdown Structure: 8 phases, 38 work packages, one named owner each.

Comment & Assessment

◁Comment and assessment by the instructor.▷